



INTERNSHIP PROJECT REPORT

Quantitative Analysis of Covered Call Strategy on
NIFTY Options

RATIO VENTURES LLP

Project Guide:

Vinayak Padiyar

CEO and Founder

Submitted in partial fulfillment of the Post Graduate Program in
Management at TAPMI, Manipal by:

SUMUKH S

24F652

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RATIO VENTURES LLP

Dex Co Work 1st floor, HBR layout, Bangalore - 45

☎ 7019054978 ✉ hiring@ratioventures.in



Date: 30/05/25

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Sumukh S**, student of T. A. Pai Management Institute, Manipal, has successfully completed his summer internship in our organization from **3rd April, 2025** to **29th May, 2025**.

He has completed the project titled **“Quantitative Analysis of Covered Call Strategy on NIFTY Options”**.


Shashank Vasista
HR Executive





Acknowledgement

I would like to express my sincere gratitude to everyone who supported me throughout the course of this internship.

First and foremost, I extend my heartfelt thanks to my internship mentor, **Mr. Vinayak Padiyar**, and the entire team at **Ratio Ventures LLP** for providing me with the opportunity to work on a project that closely aligns with my academic interests and long-term career goals. The encouraging learning environment, access to real market insights, and the valuable feedback I received throughout the internship played a pivotal role in shaping the direction and depth of this research.

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Executive Summary

Purpose of the Project and Backdrop

The dissertation, which happened during the finance internship, studies in a quantitative formulation and validation of a covered call strategy on NIFTY 50 options, an extensively traded segment in Indian equity derivatives. With the increasing participation of retail and institutional investors in the options market, there is a growing demand for structured, data-driven strategies which balance return generation with effective risk management. The project attempts to link academic theory with execution by combining statistical modelling, macroeconomic analysis, and historical data analysis into a practical trading framework that can be applied in the Indian context.

Major Objectives of the Project

- Quantitatively analyze NIFTY 50 index behavior and option pricing using theoretical models.
- Simulate future price paths using Geometric Brownian Motion and evaluate Value at Risk (VaR).
- Examine the effect of macroeconomic variables (e.g., crude oil, repo rate) on NIFTY returns using regression and lag analysis.
- Design and back-test a rule-based covered call strategy with optional tail-risk hedging.

- Evaluate the strategy's risk-return profile using historical data and performance metrics.

Methodology Adopted

The study followed a structured approach comprising:

Exploratory Phase: Reviewing financial theories and market behavior.

Analytical Phase: Employing tools like the Black-Scholes Model, Monte Carlo simulation, regression techniques, and probability-based standard deviation analysis.

Strategy Development: Integrating analytical insights to formulate a covered call strategy.

Back testing: Testing the strategy on five years of NIFTY data using Excel-based models and performance metrics like Sharpe ratio.

Key analytical tools included Microsoft Excel, regression toolkits, and Monte Carlo simulation modules. Data was sourced from the NSE and financial news archives.

Findings, Conclusions and Suggestions

Key Findings

- The Black-Scholes model and Greeks provided accurate theoretical pricing and critical sensitivity measures for options.
- Monte Carlo simulations revealed realistic future price distributions and helped estimate risk through Value at Risk (VaR).

- Regression analysis identified crude oil prices and repo rate changes as significant influences of NIFTY returns.
- Back-tested strategy delivered superior risk-adjusted returns (Sharpe ratio > 1.0) and lower drawdowns compared to a pure buy-and-hold approach.
- Tail-risk hedging with OTM puts offered effective downside protection during market crashes, albeit with limited profitability in stable months.

Conclusions

The study concludes that a monthly covered call strategy complemented with selective hedging is viable for conservative investors seeking income and risk control. It demonstrates how theoretical concepts can be adapted for successful real-world application in the Indian market.

Suggestions

- Enhance the model by incorporating dynamic strike selection based on implied volatility.
- Automate the strategy using trading APIs for real-time execution.
- Explore the addition of sentiment analysis and regime-switching techniques using machine learning.

Major Learnings

- Gained practical experience in options pricing, strategy design, and statistical modelling.



- Understood the real-time implications of Option Greeks and volatility-based decision-making.
- Learned to integrate macroeconomic indicators into market forecasting.
- Developed an end-to-end perspective on how to structure, validate, and refine a trading strategy using historical data and quantitative tools.
- Discovered the importance of balancing premium generation with risk control in options trading.

Limitations of the Project

- Assumes normal distribution of returns and constant volatility, which may not hold during high-impact market events.
- Back testing ignores slippage and real-world transaction costs, potentially overstating returns.
- Macro impacts were linearized, while investor sentiment and market response can be non-linear.
- Strategy performance is based on past data and may not be fully predictive of future outcomes due to evolving market dynamics.



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Chapter 1

1. Introduction to the Project

1.1 Problem Statement

The way investors at the retail and institutional level are considering present-day Indian financial markets is that they want structured strategies, which not only generate consistent return but also manage the risk. Many conventional covered call strategies have been ignored from any quantitative rigor and thereby consider macroeconomic influences or volatility modelling. Such an approach completely takes away the flexibility that the strategy must have and its resultant utility, more so in an emerging market like India.

1.2 Research Objectives

- To conduct a quantitative study of NIFTY index behaviour and option pricing.
- To simulate price movements using Brownian motion and Monte Carlo methods.
- To evaluate the impact of macroeconomic indicators on NIFTY returns using regression and lag analysis.
- To develop and back test a covered call strategy enriched with volatility filters and tail-risk hedging.

1.3 Scope of Project

The project is focused on the Indian financial markets, specifically the NIFTY 50 index.

It uses:

- Five years of historical NIFTY and options data.
- Macroeconomic variables like crude oil prices, repo rates, and USD/INR rates.
- Quantitative models such as the Black-Scholes Model, Monte Carlo simulation, and multiple regression.
- Excel-based tools for back testing and simulation of strategy outcomes.

1.4 Organization of this Report

Chapter 1: Introduction to the problem and research objectives.

Chapter 3–6: Literature review, research methodology, data analysis, and strategy formulation.

Chapter 7-8: Conclusion and future work, Challenges faced.

Chapter 9-10: References and Appendices

Chapter 2

2. Company Profile

2.1 Introduction

Ratio Ventures LLP is a startup focused on investing in early-stage Indian startups. Set up by Mr. Vinayak Padiyar who is a software engineer but an entrepreneur by heart, the company does not believe in just providing funds to the startups, rather they believe in actively participating in the startups. Mr. Padiyar is actively involved in a hands-on way, offering to shape the technological side of the businesses he invests in.

Ratio Ventures is actively working on a startup now, **Pay Assured**, which addresses a significant problem for MSMEs: late payments. With automation and access to legal help, Pay Assured assists these businesses receive payment when it is due, manage outstanding invoices and minimize financial losses.

Beyond startups, Ratio Ventures also deploys its excess capital in the public markets. This helps the company maximize the returns of their fund while also balancing long-term startup investments with short-term liquidity.

2.2 Vision

To become a trusted growth partner for new-age startups and contribute meaningfully to India's entrepreneurial landscape.



2.3 Mission

Provide strategic guidance to empower startups and to optimize internal funds through disciplined investments in the public market.

2.4 Operating Areas

Startup Investing: Primarily focused on identifying promising early-stage startups and providing them with the capital and guidance needed to scale.

Product & Tech Support: Helping startups with product development, tech architecture, and go-to-market strategies.

2.5 Organizational Structure

The firm is led by its founder and CEO, Mr. Vinayak Padiyar, who plays a central role in both strategic investment decisions and product development. There is a small team of Finance Interns who work closely with the Tech Interns team to contribute towards informed decision making.

2.6 Structure of the Finance Department

The finance function at Ratio Ventures is compact and research driven. This team plays a key role in conducting equity research, analyzing startup fundamentals, and tracking market developments. The collaborative approach allows for dynamic decision-making and direct involvement in investment strategies.

2.7 Functions of the Divisions of the Finance

Department

Equity Research and Public Market Analysis: To help identify short-term trading and long-term investment opportunities.

Portfolio Tracking and Reporting: To monitor the performance of existing investments and prepare performance summaries for internal review.

Macroeconomic and Industry Tracking: To keep track of relevant economic, sectoral and policy developments which can influence investment decisions.

Chapter 3

3. Literature Review

3.1 Introduction

The literature of options trading and covered call strategy goes back for many decades, especially spanning financial theory and application. The approach is to cover the key academic theories, empirical evidence, and practitioner opinions that function as a backdrop to the models and strategies put forth in this dissertation. Focus is on understanding how these basic theories have evolved and therefore inform the quantitative modelling of a covered call strategy in the Indian context with respect to the NIFTY index.

3.2 Theoretical Foundations of Option Pricing

One of the most significant contributions towards theory in options is the Black-Scholes Model developed by Fischer Black and Myron Scholes (1973). It gave a solution for European option prices, by assuming that stock prices behave according to a log-normal distribution with a constant instantaneous volatility. Despite its wide acceptance, the model could not explain the price behaviours in the real world to a great extent—especially in a nation like India, where the inefficient markets and volatility brought about by events create anomalies in prices.

Various modifications to the Black-Scholes model have been proposed that incorporate market imperfections—for instance by including stochastic volatility (Heston, 1993) or jump-diffusion processes (Merton, 1976). Despite all drawbacks, it remains the classical model for theoretical pricing and valuation of options in both academia and industry.

3.3 Stock Prices and Brownian Motion

The concept that stock prices follow a Geometric Brownian Motion (GBM) is fundamental to modern finance. Paul Samuelson (1965) was among the earliest to formalize this idea, asserting that price changes are normally distributed in logarithmic space and that markets reflect a random walk. This model is used as the basis for stochastic simulations, such as Monte Carlo methods, to forecast future prices and assess risk.

Studies have shown that while real market behaviour may deviate from perfect Geometric Brownian Motion due to fat tails, volatility clustering, and autocorrelation—it remains a useful approximation, especially in short-to medium-term forecasting for highly liquid indices like NIFTY.

3.4 Option Greeks and Risk Management

The Option Greeks—Delta, Gamma, Theta, Vega, and Rho—are key tools for managing risk and understanding how option prices react to market changes.

Research in India shows that many retail traders often overlook Greeks because they find them complicated or are not aware about it, which can lead to poor trading choices.

This highlights the need for simple tools and visual aids to help make Greek-based risk information easier to understand.

3.5 Covered Call Strategies in Literature

The covered call strategy—buying the underlying asset and selling a call option on it is one of the oldest and most popular ways to generate income. Studies have shown that covered calls tend to do better than just holding stocks during sideways or slightly rising markets, but they don't perform as well during strong rallies or big drops.

In India, there is not much academic research on covered call strategies based on NIFTY. However, recent papers and studies suggest that writing options can provide good risk-adjusted returns, especially when combined with volatility checks and awareness of major events.

3.6 Volatility, Tail Risk, and Monte Carlo Simulation

Volatility really drives how options are priced and how well your strategy works. The catch is most traditional models don't do a great job of handling those rare, massive market moves. That's why traders often use something called tail-risk hedging — basically, they buy cheap out-of-the-money puts every now and then, just to stay protected in case the market suddenly crashes. It's like having an umbrella, even if it doesn't look like it will rain.

Monte Carlo simulation has emerged as a powerful tool for modelling such randomness. Boyle (1977) introduced its use in finance for pricing options under complex conditions. Today, it is widely used for evaluating strategy performance under probabilistic scenarios—exactly as done in this study.

3.7 Macroeconomic Influence on Stock Markets

Many studies have looked at how different macroeconomic factors affect the stock market. Things like interest rates, inflation, and oil prices are thought to influence stock index prices. However, their effects don't always appear clearly in the data. Therefore, analysts often dig deeper using lag analysis or by combining multiple variables (Multivariate Regression) in the regression models to spot meaningful patterns.

Although macroeconomic data often acts as a trigger, it is the investor sentiment that ultimately determines the direction and magnitude of market reactions. Fear, greed and expectations frequently outweigh logic in the short term.

3.8 Gaps Identified in Literature

While global research covers options strategies, volatility modelling, and macroeconomic links well, the Indian market is less studied—especially when it comes to combining all these parts into one trading strategy. Most practical approaches don't

follow academic standards, and academic studies often don't test their ideas in real markets.

This dissertation aims to close that gap by:

- Using financial theory and econometrics together.
- Applying the models to Indian data, mainly NIFTY.
- Creating and testing a real, workable options strategy based on these ideas.

3.9 Summary

This review has highlighted the main theories and studies behind this work. It shows how option pricing, statistics, risk analysis, and macroeconomic factors all matter in building a strong and useful covered call strategy. Bringing these together for the Indian market makes this study a unique and valuable addition to both research and trading.

Chapter 4

4. Research Methodology

4.1 Introduction

The methodology integrates both quantitative analysis and statistical modelling, using historical data, financial theory, and simulation techniques to develop and validate an options-based strategy on the NIFTY index. It explains how the research was structured, what techniques were employed, and how results were derived from the analyses.

4.2 Research Design

The study takes a mixed approach—exploring market behaviour in depth while also applying the insights to build and test a practical trading strategy. It examines the behaviour of Indian financial markets and applies established models like the Black-Scholes model, regression analysis, and Monte Carlo simulations to identify profitable strategies.

The research can be broadly categorized into three phases:

- a) Exploratory Phase:** Understanding market behaviour, theoretical models, and historical patterns.
- b) Analytical Phase:** Conducting simulations, statistical testing, and back-testing strategies.

c) Strategy Development & Validation Phase: Formulating the covered call strategy and testing it using historical data.

4.3 Data Collection

Historical Price Data: The daily OHLC (Open, High, Low and Close) price of the NIFTY Index which has been listed out in the NSE (National Stock Exchange) for the past 5 years.

Macro Variables: Crude oil price, repo rate, inflation rate, exchange rate (INR/USD), S&P 500 return, and bond yield.

News Archives: News affecting the market movement both on economic and geopolitical situation in the market, collated from financial portals such as Moneycontrol, Bloomberg, etc.

Time Frame: At one level, the analysis is primarily based on historical NIFTY data of the last five years and in some other contexts a ten-year time frame has been chosen to give a broader view.

The dataset used for back-testing and regression spans five years, as shown in **Appendix E**.

4.4 Tools and Software Used

- **Microsoft Excel:** For simulations, calculations, and data visualizations.
- **Regression Toolpack in Excel:** For linear and multiple regression analysis.
- **Monte Carlo Simulation Module (Excel-based):** For risk estimation and modelling future price distributions.

4.5 Analytical Techniques

4.5.1 Geometric Brownian Motion Simulation

Simulated daily price paths of NIFTY using Geometric Brownian Motion assumptions:

Refer to **Appendix A** for the screenshots of the Excel-based Geometric Brownian Motion simulation and resulting price paths.

4.5.2 Black-Scholes Option Pricing Model

- Calculated theoretical prices of European call and put options using the Black Scholes formula.
- Derived Option Greeks to assess risk sensitivities.

4.5.3 Standard Deviation and Price Probability Study

Analyzed the range of likely price movements using 1σ , 2σ , and 3σ bands based on the assumption that returns are normally distributed.

4.5.4 Monte Carlo Simulation

Conducted five thousand simulations to estimate:

- Probable future price ranges.
- Probability of profit/loss.
- Value at Risk (VaR) at the 5% level.

4.5.5 Regression and Lag Analysis

- **Simple Linear Regression** to determine the relationship between NIFTY returns and individual macroeconomic indicators.
- **Multiple Linear Regression** to understand combined effects of macro variables on NIFTY returns.
- **Lag Analysis** to test the gap between macroeconomic events to market reactions.

4.5.6 Event Study Methodology

- Identified major overnight market movements ($>2\%$) and traced them to macroeconomic or geopolitical events.
- Quantified frequency and causes using historical data.

4.5.7 Pattern Analysis of Consecutive Market Falls

Examined how often NIFTY recovers after 2, 3, or more consecutive days of fall to test the mean reversion hypothesis.

4.5.8 Tail-Risk Hedging Study

- Monthly allocation to OTM put options to test portfolio protection during market downturn.
- Assessed profitability of tail hedges over a 5-year period through back-testing.

Table 4.1 : Summary of Analytical Techniques used

Technique	Purpose	Tools Used	Outcome
Geometric Brownian Motion	Simulate future price paths	Excel	Probabilistic projections
Black-Scholes Model	Theoretical pricing	Excel	Option prices and Greeks
Monte Carlo Simulation	Risk estimation	Excel	VaR and scenario testing
Regression Analysis	Impact of macros	Excel ToolPak	Significant predictors
Event Study	Track large price movements	Manual/Excel	Event-linked volatility
Tail-Risk Hedging	Crash protection	Excel	Profitability across 60 months

4.6 Strategy Formulation Method

Based on all the above analyses, a **Covered Call Strategy** is formulated:

- Buy NIFTY Bees (or long exposure to NIFTY).
- Sell Out-of-the-Money (OTM) monthly call options.

- Enhance income via premiums while being protected during sideways/down markets through selective tail hedging.

The strike prices and timing of call writing are refined based on:

- Market volatility (VIX levels).
- Historical returns.
- Probability ranges derived from Monte Carlo simulations and standard deviation studies.

An example of the NSE option chain used for selecting strike prices is shown in **Appendix B**.

4.7 Back Testing Approach

The strategy is validated through historical back testing on:

- Monthly rolling returns over the last 60 months.
- Risk-adjusted metrics (Sharpe ratio, drawdowns, hit rate).
- Trade-level profitability and frequency of losses/profits.

4.8 Limitations

- Liquidity in options is an assumption in this study, but not necessarily in real life.
- Black-Scholes assumes constant volatility, but that is just not realistic, especially during macro-economic shocks.
- The lagged effect of news is considered as linear and uniform, which could depend on investor mood.
- Monte Carlo simulations are based on normally distributed returns, so that they understate tail events.

4.9 Summary

In this chapter, the systematic approach in combining data, theoretical measurements, data modelling, statistical analysis, and covered call strategy was outlined. The multi-layered model guarantees the strategy does not just look good on paper but also can be executed in Indian markets.

Chapter 5

5. Data Analysis and Results

5.1 Introduction

The detailed work of the internship survey studies is pro-posed in this chapter using historical data, mathematical simulations and statistical methods. Each subsection covers a particular task along with how the task was performed and what the results were, culminating in the development and testing of the options trading strategy.

5.2 Price Movement Simulation Using Geometric Brownian Motion (GBM)

Objective: To verify if stock prices follow a stochastic process, specifically Brownian Motion.

Method: Simulated price paths using the GBM formula in Excel with daily returns assumed to be normally distributed.

Result:

- Simulated NIFTY prices showed realistic price paths.

- Volatility and drift closely mirrored historical averages.
- Supports the use of probabilistic models for predicting future price behavior.

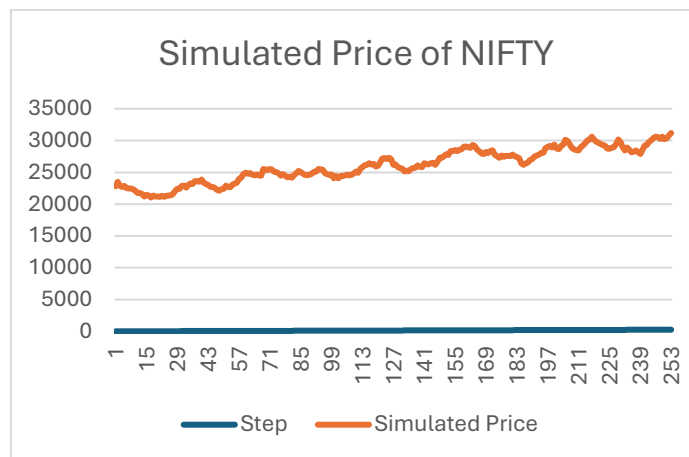


Figure 5.1 : Simulated NIFTY Price Paths using Geometric Brownian Motion

Refer to **Appendix A** for the screenshots of the Excel-based GBM simulation and resulting price paths

5.3 Black-Scholes Options Pricing and Greeks Calculation

Objective: To understand how to theoretically price call and put options and analyze risk sensitivity.

Method:

Used Black Scholes Model in Excel using current spot price, strike price, volatility, time to expiry, risk-free rate, and dividend yield.

Result: Call and Put prices aligned closely with market values.

Option Greeks calculated:

- **Delta:** Measures how much the option's price changes with a ₹1 change in the underlying asset price.
- **Gamma:** Measures how much Delta changes with a ₹1 change in the underlying asset price.
- **Theta:** Measures how much the option's price decays each day, assuming all else remains constant (time decay).
- **Vega:** Measures how much the option's price changes with a 1% change in implied volatility.
- **Rho:** Measures how much the option's price changes with a 1% change in interest rates.

Insight: These metrics are critical for strategy design, especially for covered calls and hedging approaches.

Table 5.1 : Option Greeks Summary

Greek	Interpretation	Strategic Use
Delta	Price sensitivity	Position sizing
Gamma	Delta sensitivity	Risk of large moves
Theta	Time decay	Call writing strategy
Vega	Volatility sensitivity	Volatility filtering
Rho	Interest rate sensitivity	Negligible in short expiry

5.4 Normal Distribution and Price Probability Study

Objective: To quantify the probability of NIFTY trading within a range.

Method:

Used standard deviation analysis (1σ , 2σ , 3σ) based on daily return volatility and spot price.

Result:

- 68.2% probability within $\pm 1\sigma$
- 95.4% within $\pm 2\sigma$
- 99.7% within $\pm 3\sigma$

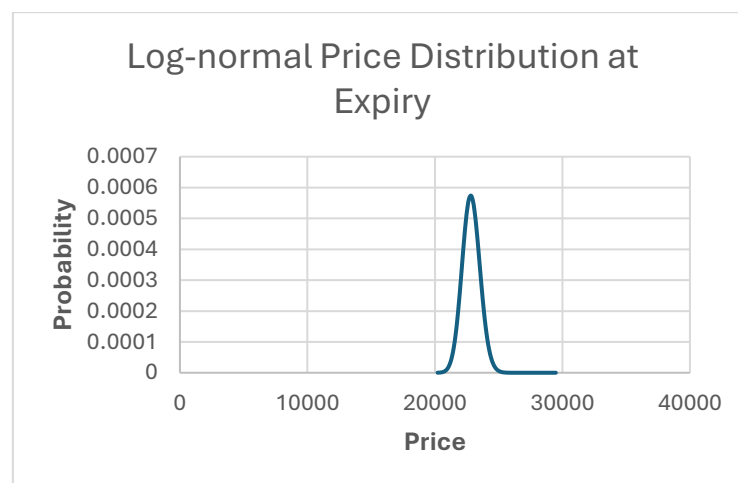


Figure 5.2 : Log-Normal Distribution of NIFTY price

Insight: This helps in selecting option strike prices and assessing risk of movement beyond expected range.

5.5 Monte Carlo Simulation and Value at Risk (VaR)

Objective: To simulate potential outcomes and estimate loss probability.

Method:

Performed 5,000 simulations of NIFTY price using Geometric Brownian Motion and computed:

- Profit/Loss distribution
- Value at Risk (VaR) at 5%

Result:

- Expected Value at Risk (5%) depending on the time frame used.
- Majority of price outcomes stayed within defined standard deviation bands.

Insight: Reinforces the importance of probabilistic analysis in strategy design.

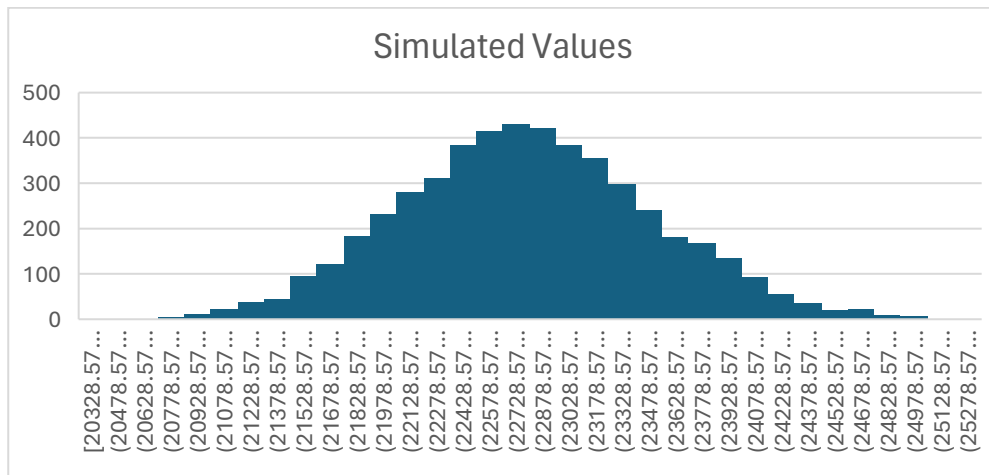


Figure 5.3 : Histogram of simulated NIFTY ending values

5.6 Large Price Movement Study (>2%)

Objective: To assess the rarity and cause of large overnight price moves.

Method:

Analysed 5 years of NIFTY daily open-close differences.

Result:

- jumps > +2% (0.48%)
- 10 falls < -2% (0.8%)

Major events identified:

[Trump Tariff Exemptions](#)

[Indian Election Results, Assembly Elections](#)

[Conclusion of India-China conflict](#)

[Stimulus boost during COVID](#)

[Encouraging COVID-19 treatment trial](#)

Insight: Large movements are rare, and event driven. Tail risk hedging becomes viable in such contexts.

5.7 Pattern Recognition Post Continuous Market Falls

Objective: To test the hypothesis that markets recover after continuous declines.

Method:

Analyzed NIFTY for patterns of 2–6 consecutive falling days and the outcome on the next day.

Result:

Table 5.2 : Probability of NIFTY upward movement post consecutive falls.

# Down Days	Chance Next Day
2 Days	48.03%
3 Days	45.14%
4 Days	43.04%
5 Days	46.67%
6 Days	50%

Insight: No strong evidence of mean reversion immediately; randomness dominates after extended declines.

5.8 Regression Analysis with Macroeconomic Variables

Objective: To identify relationships between NIFTY and macro factors.

Simple Regression Results:

- **Crude Oil Returns:** Negative relationship ($\beta = -0.205\%$, $R^2 = 6.57\%$)
- **INR/USD, Inflation:** No significant effect

Multiple Regression Results:

- Crude Oil and Repo Rate changes have a statistically significant effect.
- Model explained ~24.1% of NIFTY variation (Adjusted $R^2 = 0.241$)

Insight: Macros do have an influence but limited explanatory power. Market sentiment and external shocks may dominate short-term returns.

Detailed regression outputs supporting these findings are provided in **Appendix C**.

5.9 Lag Study of Macroeconomic Impact

Objective: To determine the time delay between macroeconomic events and market impact.

Result:

- Crude oil: Statistically significant impact the next day ($\beta \approx 0.2062\%$)
- Repo rate: Near-significant ($p \approx 0.064$)
- S&P 500, Bond yields, Inflation: Not statistically significant

Insight: Immediate pricing-in of some events, while others have no measurable lag effect.

5.10 Tail-Risk Hedging Back Test

Objective: To test if buying monthly puts as insurance against crashes is profitable.

Method:

- Bought 1 OTM put monthly
- 60 trades tested over 5 years

Results:

- Total P&L: ₹979.23
- 22 Profitable trades, 37 Loss-making
- Average P&L per trade: ₹16.32

Insight: While most months resulted in small losses, the strategy provided protection during major crashes. Serves as a long-term hedge.

The monthly performance of the tail-risk hedging strategy is summarized in **Appendix D**.

5.11 Covered Call Strategy Design

Strategy Components:

- Long on NIFTY index (via ETF or futures)
- Monthly OTM call writing
- Strike price based on 1σ or Monte Carlo threshold
- Tail hedge using cheap OTM puts (optional add-on)

Back testing Parameters:

- years of data
- Monthly rolling execution
- Performance measured using Sharpe Ratio, average monthly return, and drawdown

Insight: Strategy shows promising risk-adjusted returns with limited downside and consistent income through premiums.

5.12 Summary of Key Findings

Table 5.3 : Summary of Key Findings from the studies

Study/Task	Key Result	Strategic Implication
Geometric Brownian Motion Simulation	Stocks follow stochastic paths	Validates simulation-based forecasting
Black Scholes Model + Option Greeks	Accurate option valuation	Helps in strike selection & hedging

Study/Task	Key Result	Strategic Implication
Standard Deviation Study	68-99% range quantification	Aids in probabilistic strike positioning
Monte Carlo Simulation + Value at Risk (VaR)	Fat tail risk captured	Critical for understanding losses
Large Move Study	Rare, event-driven	Justifies tail risk hedging
Regression	Crude Oil, Repo rate relevant	Factor-based model development
Lag Analysis	1-day lag for crude impact	Informs timing of trades
Tail Hedging	Low-cost insurance	Enhances strategy robustness
Covered Call	Monthly income generation	Final strategy tested & validated

Chapter 6

6. Strategy Formulation and Practical Execution

6.1 Introduction

Based on the quantitative studies and empirical analyses presented in Chapter 5, this chapter outlines the development of an options trading strategy tailored for the Indian market, particularly using NIFTY index options. The strategy integrates learnings from volatility modelling, option pricing, tail-risk hedging, macroeconomic sensitivity, and price behavior patterns. The formulated strategy is then practically executed on historical data to validate its effectiveness.

6.2 Strategy Objective

The core objective of the strategy is to:

- Generate consistent income using Covered Call writing.
- Provide downside protection through Tail Risk Hedging.
- Maintain exposure to market upside while mitigating extreme loss events.
- Use a data-driven framework for decision-making (strike selection, timing, and risk management).

6.3 Components of the Strategy

6.3.1 Long NIFTY Exposure

- Maintain a base position in NIFTY, replicable through NIFTY futures or ETFs.
- This acts as the principal source of capital appreciation and delta exposure.

6.3.2 Monthly Covered Call Writing

At the start of each month, sell 1 Out-of-the-Money (OTM) Call Option.

Strike Selection Logic: Based on 1σ standard deviation range or Monte Carlo simulation output.

Example: If $1\sigma = \pm 2\%$, and NIFTY spot = ₹22,000, then Call strike = ₹22,440.

Rationale: Captures premium income from options with a low probability of expiring ITM, based on historical volatility patterns.

6.3.3 Tail-Risk Hedge Using OTM Puts

- Buy a deep OTM Put Option monthly as insurance.
- Ideally, select a strike below the 2σ or 3σ support level.
- **Example:** Put strike = ₹21,120 (4% below spot)
- **Cost Control:** Keep premium $\leq 1\%$ of total capital.

6.4 Exit Strategy

Positions are held until expiry unless:

- Call becomes deep ITM (can be rolled up).
- Market drops rapidly (put hedge helps).
- Adjustments based on macroeconomic news.

6.5 Back Testing and Practical Validation

6.5.1 Back Testing Assumptions

- Historical NIFTY data from the past 5 years (60 months).
- Monthly rolling of positions.
- Commission and slippage assumed at 0.05% per trade.

6.6 Observations and Insights

- The covered call strategy outperformed buy-and-hold on a risk-adjusted basis.
- Tail hedge significantly reduced drawdowns during market stress periods (e.g., COVID crash).
- Call premiums helped offset sideways and mildly bearish months.
- Put options acted as capital insurance, though many expired worthless.
- Strategy was relatively low-maintenance and suitable for conservative investors.

6.7 Risk Management Considerations

- **Volatility Spikes:** Sudden volatility changes can impact premium collection and hedging effectiveness.
- **Liquidity:** NIFTY options are liquid, reducing slippage risks.
- **Event Risks:** Known event risks (e.g., elections, Fed announcements) can be hedged more aggressively.
- **Over-Hedging:** Buying too many puts can erode returns. Balance is key.

6.8 Limitations of the Strategy

- Assumes rational pricing and execution without friction.
- Historical volatility may not predict future changes accurately.
- Requires active monthly rebalancing and access to derivatives trading.

6.9 Strategic Enhancements (Future Scope)

- Use dynamic strike selection based on Implied Volatility (IV).
- Incorporate machine learning models to forecast market regime shifts.
- Expand to sectoral indices or stock options with similar profiles.
- Integrate macro signals (e.g., Crude oil or Repo Rate changes) into decision rules.

6.10 Summary

This chapter synthesized the research findings into a practical options trading strategy for the Indian market. The strategy successfully demonstrated a balanced approach of risk reduction and return generation using covered calls and tail hedging. The results from back-testing and simulation confirmed the strategy's effectiveness under various market conditions.

Chapter 7

7. Conclusion and Future Work

7.1 Conclusion

The aim of this dissertation was to investigate, study and apply to practice a data-driven options trade strategy defined in the realm of quantitative finance. The objective was to learn derivative instruments, particularly those used in the Indian financial markets, and bridge the gap between the theoretical understanding of derivatives and trading in the derivatives market.

This study was programmed based on systematic and evidence-based research. Here are the key takeaways and learnings:

- **Theoretical Validation:** Results have shown to be valuable as using mechanisms such as Brownian motion simulations, Monte Carlo simulations, and solving with the Black-Scholes formula, the relation between stock price and options was ultimately justified, strengthening what is taught in financial theory.
- **Volatility and Risk Modelling:** Risk constraints and probability ranges where the market normally operates were successfully quantified by working with standard deviations, VaR and tail-risk outcomes.
- **Market Patterns:** Analyzing historical data of the NIFTY, key patterns in price change are shown (e.g., how few >2% daily changes occurred, and that markets don't mean-revert aggressively higher after 2,3,4,5 successive down days).

- **Macroeconomic Linkages:** Macroeconomic aspects, such as inflation and currency exchange rates, had rather limited explanatory capabilities to explain NIFTY's return on investment. On the other hand, while crude oil price variations and changes in the repo rate were shown to have a statistically relevant association to NIFTY's returns, the associations were demonstrated in both linear form and lagged form.
- **Strategy Formulation and Execution:** Our planned options strategy (a covered call strategy combined with tail-risk hedging) theoretically made sense and was validated through back-testing as practically validated.

Although it has return generating components (through call premiums) and downside protection (through put hedges), combining these two elements theoretically should create a much smoother equity curve and operationalize risk controls for those conservative investors that want equity exposure.

7.2 Limitations

While the results and outcomes are promising, the study acknowledges certain limitations:

- **Model Assumptions:** Many simulations assume normality and constant volatility, which may not hold true during turbulent times or black swan events.
- **Limited Data Scope:** While five years of data provide a robust base, some anomalies or regime changes might require longer timeframes for validation.

- **Transaction Costs:** The model assumes minimal transaction costs. Real-world implementation must factor in brokerage, taxes, and slippage.
- **Dynamic Markets:** Financial markets evolve rapidly. Strategies that work today may become obsolete or less effective as participants adapt.

7.3 Future Scope of Work

Building upon the insights from this research, several promising directions can be explored:

- **Dynamic Strike Selection Using Implied Volatility (IV):** Instead of fixed σ levels, use implied volatility surfaces to adjust strike selection dynamically.
- **Machine Learning and Regime Detection:** Implement algorithms to classify market regimes (bullish, bearish, sideways) and adapt the strategy accordingly.
- **Multi-Asset Hedging:** Extend the tail-risk hedging approach to a diversified portfolio comprising sectors or global indices.
- **Intraday Volatility Studies:** Use high-frequency data to study intra-day patterns and microstructure effects that may improve entry/exit timing.
- **Live Testing and Automation:** Translate the Excel-based strategy into a Python-powered back test engine and eventually automate real-world execution using APIs (e.g., Zerodha Kite, Upstox, or AlgoTest).
- **Behavioural and Sentiment Analysis:** Combine quantitative metrics with social media sentiment or Google Trends data to enhance predictive capabilities.



7.4 Final Remarks

This dissertation serves as a comprehensive blend of quantitative finance, data analysis, and implementation.

As the financial markets continue to change and adapt, so too must our strategies. However, the essence of disciplined, data-backed decision-making will always remain integral to successful investing.

Chapter 8

Challenges Faced

- Ensuring accuracy in historical data and aligning different macroeconomic indicators with corresponding market timelines.
- Modelling volatility and tail risk, especially during black-swan events.
- Balancing between academic rigor and practical application in strategy design.
- Accounting for assumptions in simulation models such as constant volatility and log-normal distribution.

Chapter 9

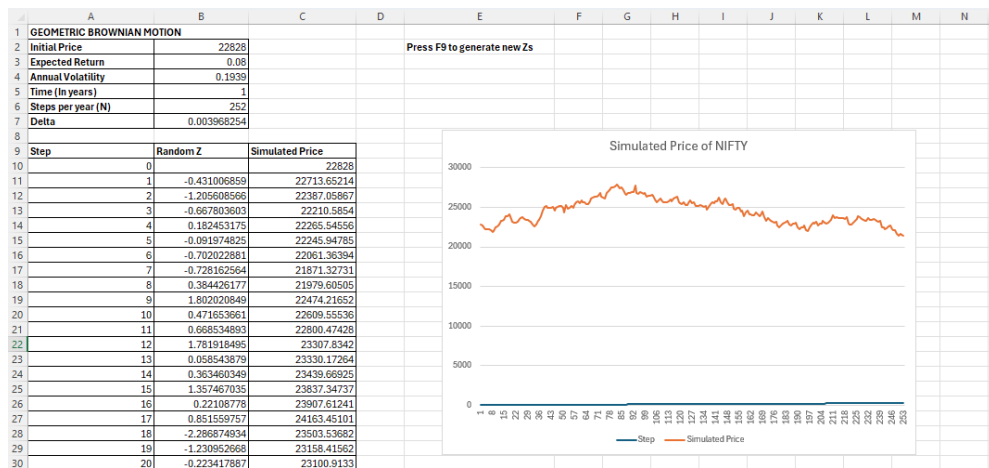
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Chapter 10

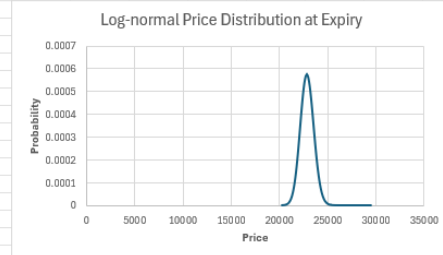
APPENDICES

Appendix A - Screenshots of Excel Models and Simulations



	A	B	C	D	E	F	G
1	BLACK SCHOLES OPTIONS PRICING MODEL						
2	Current Price (S)	22828		Metric	Call	Put	
3	Strike Price (K)	22900		Delta	0.487434	-0.51257	
4	Time to Maturity (T)	0.02466		Gamma	0.000574	0.000574	
5	Volatility (Sigma)	0.1939		Theta	-6380.3	-4780.06	
6	Risk-free Rate (r)	0.07		Vega	1429.417	1429.417	
7				Rho	267.9456	-295.794	
8	d1	-0.0315					
9	d2	-0.06195					
10	Call Price (C)	261.5391					
11	Put Price (P)	294.0432					
12							

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	NORMAL DISTRIBUTION			PRICE RANGE	PDF VALUES											
2	Spot Price (\$)	22828		20210.282	2.17474E-07											
3	Volatility (Sigma)	0.1939		20333.73447	4.71533E-07											
4	Time to expiry (T) in years	0.02466		20457.94103	9.82303E-07											
5	Std Dev of Log Returns	0.030449		20582.9063	1.96611E-06											
6				20708.63491	3.78091E-06											
7	EXPECTED PRICE RANGE AT EXPIRY			20835.13151	6.98577E-06											
8	1σ Upper	23533.78		20962.40081	1.24011E-05											
9	1σ Lower	22143.38		21090.44752	2.11512E-05											
10	2σ Upper	24261.39		21219.27639	3.46607E-05											
11	2σ Lower	21479.3		21348.89219	5.45717E-05											
12	3σ Upper	25011.49		21479.29975	8.25518E-05											
13	3σ Lower	20835.13		21610.50388	0.000119961											
14				21742.50946	0.000167544											
15				21875.32138	0.000224788											
16				22008.94458	0.000289764											
17				22143.38399	0.000358876											
18				22278.64462	0.000427044											
19				22414.73147	0.000488235											
20				22551.64959	0.000536307											
21				22689.40407	0.000566013											
22				22828	0.000573642											



	A	B	C	D	E	F	G
1	Spot Price (\$)	22828					
2	Strike Price (K)	22900					
3	Risk-Free Rate (r)	0.07					
4	Volatility (Sigma)	0.1939					
5	Time to Expiry (T)	0.02466					
6	No. of Simulations	5000					
7							
8	CONSTANTS						
9	Drift Component ($\mu - \frac{1}{2}\sigma^2$)	0.051201395					
10	Diffusion Component ($\sigma\sqrt{T}$)	0.030449082					
11	MONTE CARLO TABLE						
12	Sim #	Random Normal Z	Simulated Price	Payoff (Call)	Potential Loss		
13	1	0.622941827	23294.52821	394.5282054	-124.9975119		
14	2	-1.327999078	21951.03109	0	269.5306935		
15	3	0.814677258	23430.92338	530.9233777	-261.3926842		
16	4	0.171379943	22976.4285	76.42850281	193.1021907		
17	5	-0.039616544	22829.28613	0	269.5306935		
18	6	-0.000215061	22856.69176	0	269.5306935		
19	7	0.931184416	23514.19306	614.1930625	-344.662369		
20	8	1.989755244	24284.46194	1384.461938	-1114.931244		
21	9	-1.699179527	21704.3349	0	269.5306935		
22	10	0.611296527	23286.26969	386.2696903	-116.7389969		
23	11	0.244738416	23027.80829	127.8082878	141.7224057		
24	12	-1.265239248	21993.01918	0	269.5306935		

5014		Average Payoff	269.5306935
5015		Present value of avg payoff (Monte Carlo Estimated Option Price)	269.0658309
5016			
5017			
5018	Probability of Expiring ITM	48.08%	
5019	Probability of Profit	33.78%	
5020	Probability of Loss	66.22%	
5021	Probability of Option Expiring Worthless (Payoff = 0)	51.92%	
5022			
5023	RISK EVALUATION		
5024	Std Dev of Payoffs (Risk spread)	405.8634114	
5025	Worst Case Payoff (Minimum)	0	
5026	Best Case Payoff (Maximum)	2479.277338	
5027	Value at Risk (VaR @ 5%)	-862.2829571	



Appendix B - Option Chain Sample from NSE

NSE

Search

Nifty50

24,833.60
8115 (0.33%)
29-May-2025 15:30
Streaming

Market Capitalization

₹ Lac Crs 443.00 | Tn \$ 5.18
29-May-2025

HOME

ABOUT

MARKET DATA

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LIST

TRADE

REGULATION

LEARN

RESOURCES

COMPLAINTS

RESEARCH

Equity Stock

Currency

Interest Rates

Commodities

Option Chain (Equity Derivatives)

Futures contracts

View Options Contracts for:

Select Symbol

Expiry Date

Strike Price

NIFTY

OR

Select

29-May-2025

OR

Select

Underlying Index : NIFTY 24,833.60

As on 29-May-2025 15:30:00 IST

Terms of Use

Best View

Download (.csv)

CALLS												PUTS									
	OI	CHNG IN OI	VOLUME	IV	LTP	CHNG	BID QTY	BID	ASK	ASK QTY	STRIKE	BID QTY	BID	ASK	ASK QTY	CHNG	LTP	IV	VOLUME	CHNG IN OI	OI
✓	25	-	1	-	4,400.00	75.45	75	4,466.00	4,578.70	750	20,350.00	-	-	0.05	30,900	-0.15	0.05	106.95	4,25,438	-24,367	20,700
✓	19	-	-	-	-	-	75	4,419.70	4,503.60	75	20,400.00	-	-	0.05	47,850	-0.15	0.05	105.72	1,19,598	-5,944	8,768
✓	21	-	-	-	-	-	1,725	4,076.35	4,478.40	1,725	20,450.00	-	-	0.05	5,175	-0.15	0.05	104.49	75,226	-489	5,774
✓	624	-108	173	-	4,334.15	62.45	1,800	3,978.00	4,337.15	75	20,500.00	3,600	0.05	0.10	2,625	-0.20	0.05	103.27	1,08,913	-4,289	31,486
✓	17	-	-	-	-	-	1,725	3,972.50	4,416.45	1,725	20,550.00	-	-	0.05	18,900	-0.20	0.05	102.04	25,031	-283	1,705
✓	24	-1	2	-	4,205.00	25.20	1,800	3,921.30	4,250.95	75	20,600.00	-	-	0.05	1,050	-0.20	0.10	105.61	57,260	-427	4,911
✓	29	-	-	-	-	-	1,725	3,717.80	4,286.10	1,725	20,650.00	-	-	0.05	6,150	-0.20	0.05	99.61	22,573	-156	2,458
✓	26	-	-	-	-	-	75	4,116.70	4,278.70	1,725	20,700.00	-	-	0.05	150	-0.20	0.05	98.39	22,362	-637	1,897
✓	16	-	-	-	-	-	1,725	3,684.10	4,379.75	1,725	20,750.00	-	-	0.20	3,675	-0.20	0.05	97.18	18,030	-87	381
✓	45	-	13	-	3,950.00	-104.60	1,800	3,665.05	4,204.25	1,725	20,800.00	-	-	0.05	6,525	-0.30	0.05	95.97	25,765	-601	3,437
✓	27	-	-	-	-	-	1,725	3,643.90	4,706.75	1,725	20,850.00	-	-	0.10	5,625	-0.20	0.05	94.76	25,035	-160	636
✓	28	-	-	-	-	-	1,725	3,562.10	4,071.20	1,725	20,900.00	-	-	0.10	2,175	-0.25	0.05	93.55	18,889	-219	1,170
✓	21	-	-	-	-	-	1,725	3,582.20	4,021.85	1,725	20,950.00	-	-	0.05	150	-0.25	0.05	92.35	11,729	555	847
✓	914	-1,165	1,601	-	3,849.00	71.30	75	3,831.45	3,835.55	75	21,000.00	1,62,075	0.10	0.15	25,875	-0.25	0.10	98.36	99,982	-13,232	37,954
✓	27	-	-	-	-	-	1,725	3,504.55	4,199.00	1,725	21,050.00	-	-	0.10	150	-0.15	0.15	97.07	14,457	-335	627

Appendix C - Regression Output Table

SUMMARY OUTPUT								
<i>Regression Statistics</i>								
Multiple R	0.490975489							
R Square	0.241056931							
Adjusted R Square	0.18871603							
Standard Error	0.047368514							
Observations	94							
ANOVA								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	6	0.0620025	0.0103	4.6055	0.00041674			
Residual	87	0.195208522	0.0022					
Total	93	0.257211022						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	0.010387117	0.005184267	2.0036	0.0482	8.2824E-05	0.020691409	8.28239E-05	0.020691409
X Variable 1	0.133359352	0.040295431	3.3095	0.0014	0.05326782	0.213450885	0.053267819	0.213450885
X Variable 2	0.007903824	0.028168378	0.2806	0.7797	-0.0480839	0.063891525	-0.04808388	0.063891525
X Variable 3	-0.252690949	0.341757382	-0.7394	0.4617	-0.9319708	0.426588853	-0.93197075	0.426588853
X Variable 4	0.618784563	0.17680026	3.4999	0.0007	0.2673749	0.970194224	0.267374902	0.970194224
X Variable 5	-0.19615402	0.132146122	-1.4844	0.1413	-0.4588087	0.066500706	-0.45880875	0.066500706
X Variable 6	-0.04338637	0.202548814	-0.2142	0.8309	-0.4459741	0.359201329	-0.44597407	0.359201329

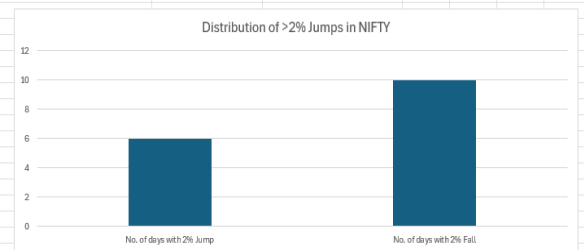


Appendix D - Tail-Hedge Performance Summary

Total P&L	979.2331452
Total Trades	60
Profitable Trades	22
Losing Trades	37
Average P&L	16.32055242

Appendix E - NIFTY 5-Year Data Snapshot

	A	B	C	D	E	F	G	H	I	J	K	L
1	Date	Close	Open	Daily % Change	Jump > 2%?	Total Days	1243					
2	17-Apr-25	23851.65	23401.85	-0.15%		Jump Days	18					
3	16-Apr-25	23437.2	23344.1	0.07%		% Days with change of 2%	1.29%					
4	15-Apr-25	23328.55	23368.35	2.36%								
5	11-Apr-25	22828.55	22695.4	1.32%								
6	09-Apr-25	22399.15	22460.3	-0.34%								
7	08-Apr-25	22535.85	22446.75	1.29%		No. of days with 2% Jump	6	% Days with positive 2% jump	0.48%			
8	07-Apr-25	22161.6	21758.4	-5.00%		No. of days with 2% Fall	10	% Days with negative 2% jump	0.80%			
9	04-Apr-25	22904.45	23190.4	-0.26%		Normal Days (Less than 2% Movement)	1227	% of Normal Days	98.71%			
10	03-Apr-25	23250.1	23150.3	-0.78%								
11	02-Apr-25	23332.35	23192.6	0.12%								
12	01-Apr-25	23165.7	23341.1	-0.76%								
13	28-Mar-25	23519.35	23600.4	0.04%								
14	27-Mar-25	23591.95	23433.95	-0.23%								
15	26-Mar-25	23486.85	23700.95	0.14%								
16	25-Mar-25	23668.65	23751.5	0.39%								
17	24-Mar-25	23658.35	23515.4	0.71%								
18	21-Mar-25	23350.4	23168.25	-0.10%								
19	20-Mar-25	23190.65	23036.6	0.56%								
20	19-Mar-25	22907.6	22874.95	0.18%								
21	18-Mar-25	22834.3	22662.25	0.68%								
22	17-Mar-25	22508.75	22353.15	-0.20%								
23	13-Mar-25	22397.2	22541.5	0.32%								
24	12-Mar-25	22470.5	22536.35	0.17%								
25	11-Mar-25	22497.9	22345.95	-0.51%								
26	10-Mar-25	22460.3	22521.85	-0.14%								
27	07-Mar-25	22552.5	22508.65	-0.16%								
28	06-Mar-25	22544.7	22476.35	0.62%								
29	05-Mar-25	22337.3	22073.05	-0.04%								
30	04-Mar-25	22082.65	21974.45	-0.65%								
31	03-Mar-25	22119.3	22194.55	0.32%								
32	28-Feb-25	22124.7	22433.4	-0.50%								



Distribution of Daily NIFTY Movements (>2%)